

Piura crop classification — what was built, what was tested, and where the project stands

This records the modelling work: the 12-class crop classifier, the move to a 3-class land-state classifier, the multi-year panel, and the validation gates it failed. Supporting documents:

[PIPELINE.md](#) (12-class pipeline), [perennial/RESULTS.md](#) (3-class detail), and [perennial/plan.md](#) (blueprint).

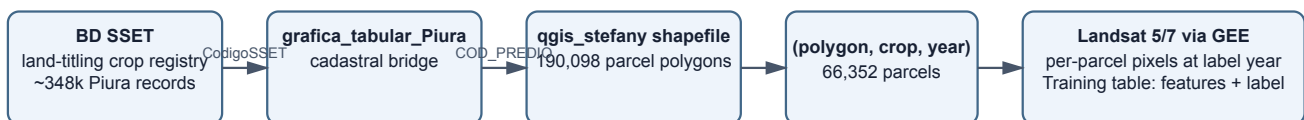
Last updated 2026-08-06.

0. Summary

The research question is “**has land in Piura shifted from domestic annual crops to export perennial crops?**” A per-parcel classifier was trained on land-titling crop records matched to Landsat imagery, then applied to a 28-year satellite panel.

The single-year classifier works: the 3-class model reaches locked-test macro-F1 **0.681**. **The 28-year panel does not yet work:** it failed temporal-transfer and trajectory-stability gates under two different model architectures. The largest problem is the 1996–98 baseline, which overlaps the catastrophic 1997–98 El Niño. **No trend has been produced and none should be reported.** The perennial signal is detectable; annual parcel-by-parcel change is not.

1. Data overview



The labels are a **single snapshot**: each parcel's crop was recorded when it was titled, around 80% in 1998–99. There is no later observation for the same parcel. Thus the data train a “what is this parcel now?” model, not a change-detection model. Detecting change requires applying the model to other years and testing whether those predictions transfer.

Titling year, geography, and crop are also entangled: 1998 parcels are 0.4% perennial, while 2000 parcels are 33.8% perennial. The most common label years also coincide with the 1997–98 El Niño, the most abnormal imagery in the record.

2. Model 1 — the 12-class crop classifier

2.1 Inputs and evaluation

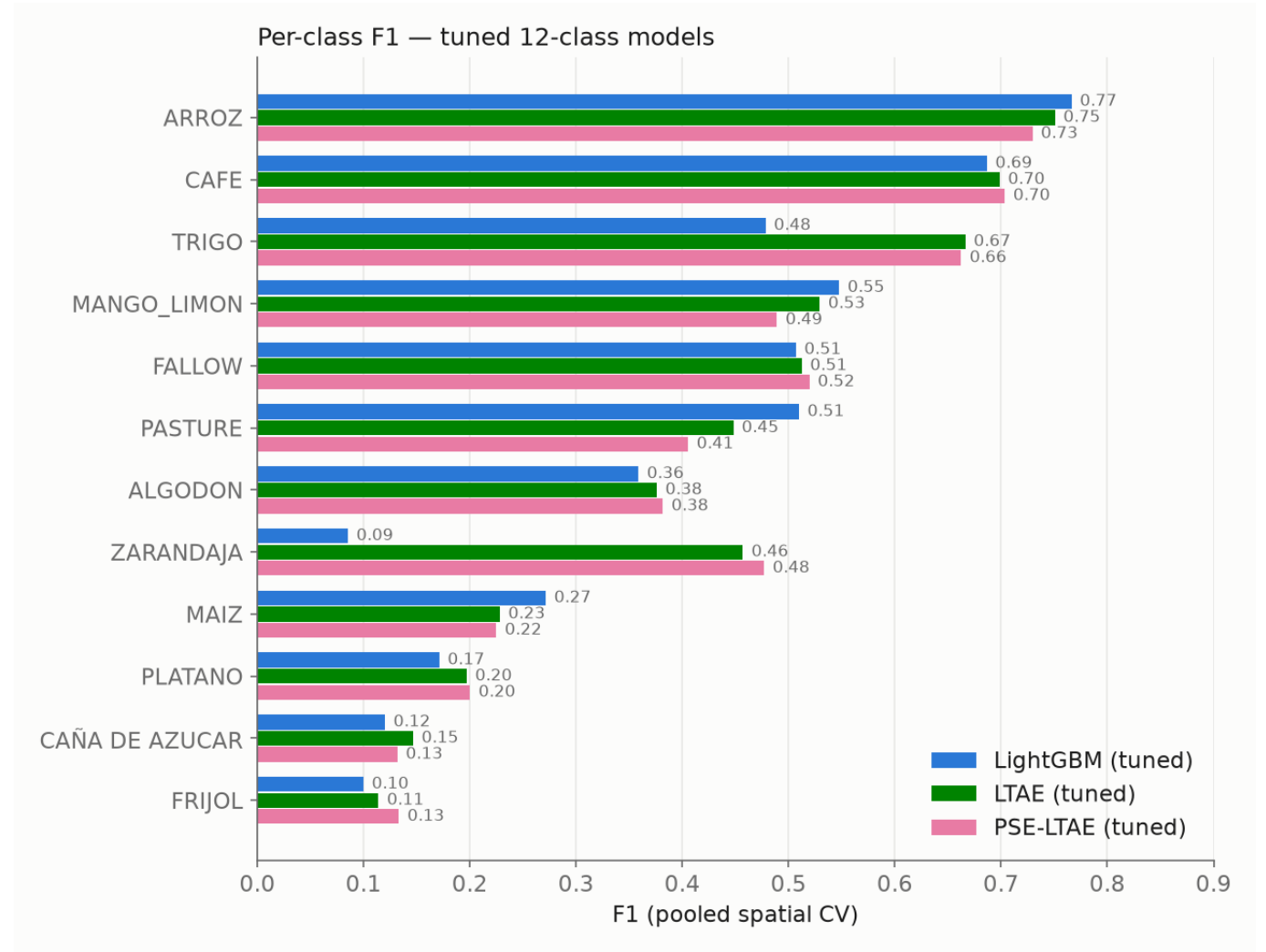
model	input	what it sees
LightGBM	flat — 140 columns	Whole-year parcel summaries from 6 Landsat bands and 5 indices, plus parcel area, latitude, and acquisition metadata.
LTAE	sequence — [N, 64, 11]	Per-date parcel-median reflectance, with day-of-year and a validity mask.
PSE-LTAE	pixelset — [N, 64, 8, 11]	The same time series, retaining 8 individual pixels per date for within-parcel texture.

The 11 channels are six Landsat bands (B G R NIR SWIR1 SWIR2) and five indices (NDVI EVI NDWI NDMI BSI). Splits are spatially blocked by contiguous 5 km regions with a 1.5 km buffer: adjacent Piura parcels often grow the same crop, so random splits would leak neighbouring farms between training and validation.

2.2 Results

12 crop classes, 49,648 parcels, pooled spatial-CV results:

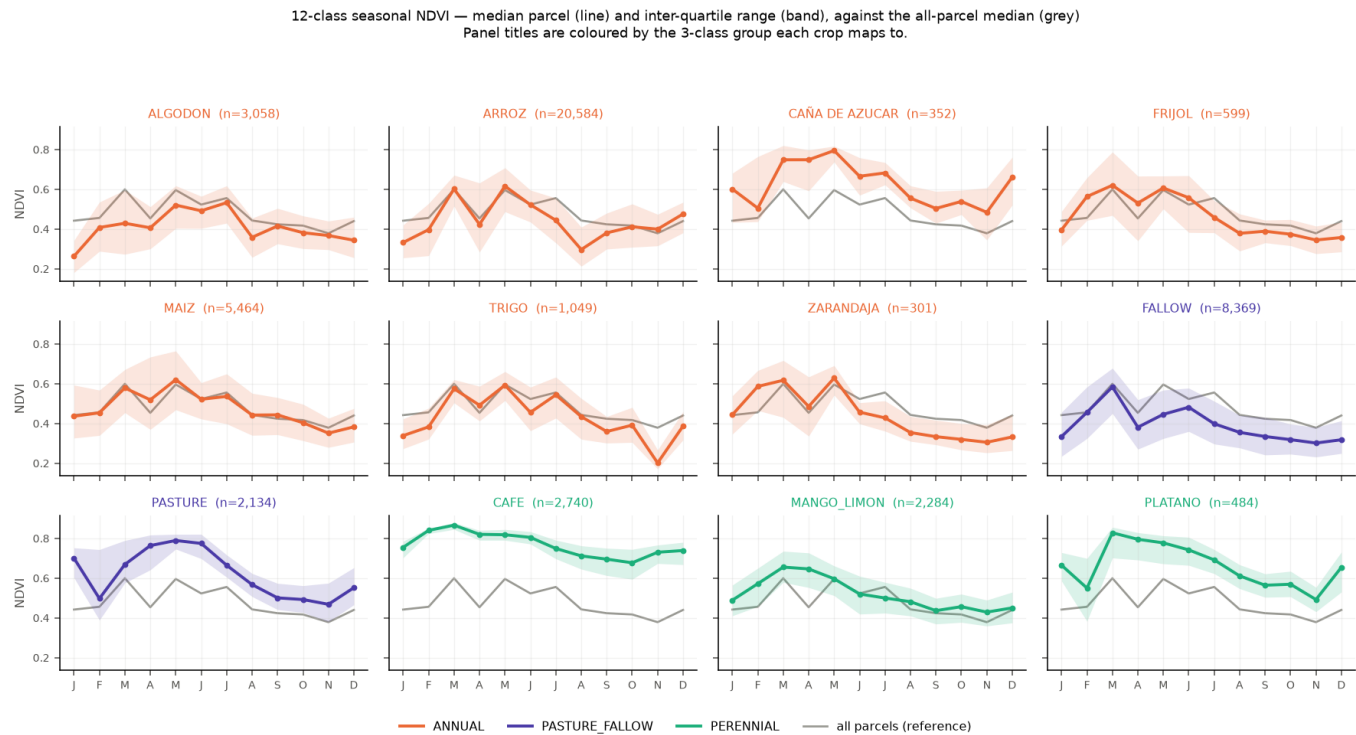
model	pooled macro-F1	accuracy
LTAE	0.427	0.559
PSE-LTAE	0.421	0.540
LightGBM	0.383	0.576



class	F1	support	class	F1	support
ARROZ (rice)	0.751	16,446	PASTURE	0.448	1,980
CAFE (coffee)	0.698	2,322	ALGODON (cotton)	0.376	2,855
TRIGO (wheat)	0.666	661	MAIZ (maize)	0.228	4,175
MANGO_LIMON	0.529	1,435	PLATANO (banana)	0.197	448
FALLOW	0.512	7,224	CAÑA DE AZUCAR	0.147	339
ZARANDAJA	0.457	296	FRIJOL (beans)	0.114	501

2.3 Why the model moved to three classes

The annual crops are not reliably separable. Maize, beans, cotton, and sugarcane have similar whole-year 30 m profiles, while rice is distinctive because it floods and coffee is distinctive because it keeps evergreen canopy.



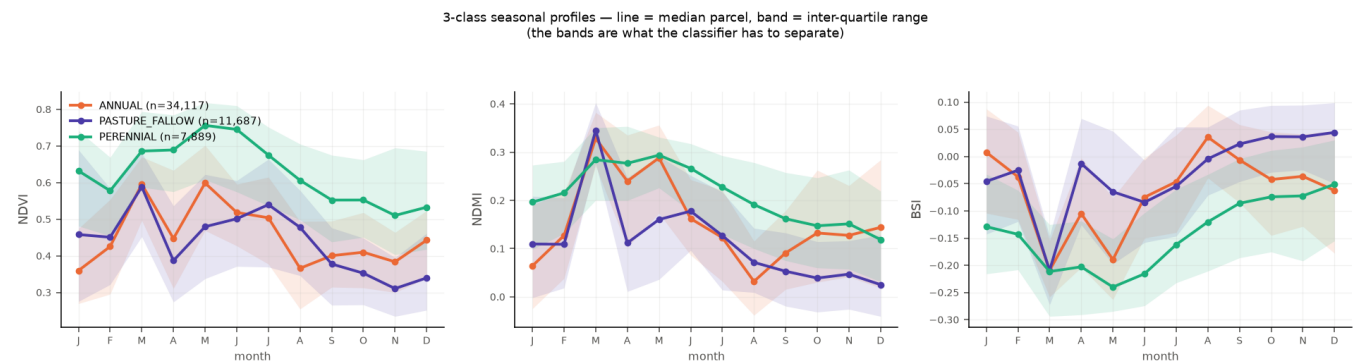
The useful physical distinction is therefore not crop species, but **whether a parcel retains canopy year-round**. That is also closer to the research question: perennial crops broadly represent export-oriented production and annual crops domestic production. This motivates the 3-class model below. (CAÑA DE AZUCAR remains grouped as annual despite its perennial-looking profile; that choice can be revisited.)

3. Model 2 — the 3-class land-state classifier

The same pipeline and spatial splits were used, but labels were regrouped as PERENNIAL, ANNUAL, and PASTURE_FALLOW. This retains 56,419 parcels, including rare or mixed crops that were unsuitable for the crop-level classifier.

class	parcels	share
ANNUAL	36,102	64.0%
PASTURE_FALLOW	12,218	21.7%
PERENNIAL	8,099	14.4%

3.1 Class separability

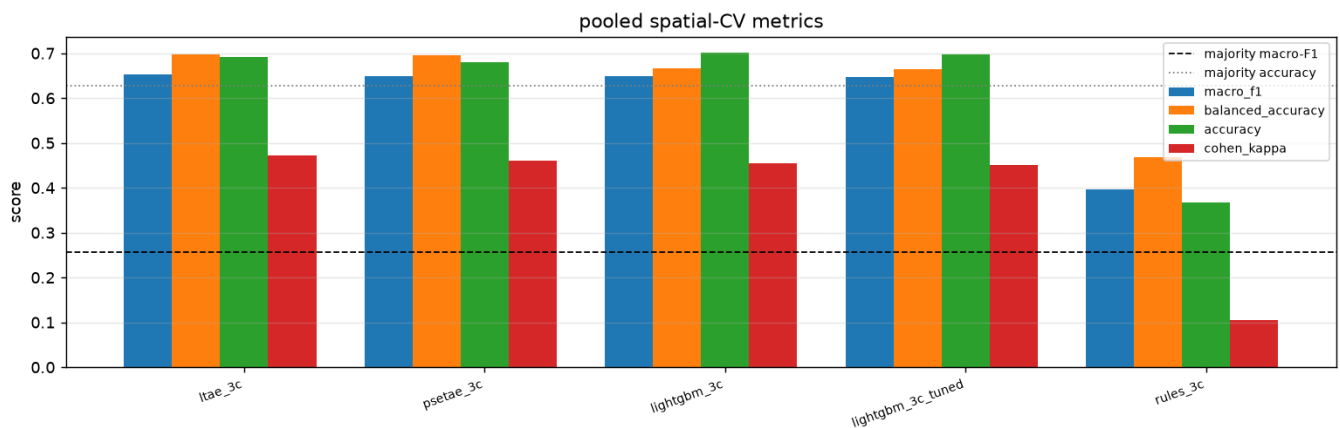


The classes separate in the expected direction. In Aug–Dec, perennial parcels retain NDVI around 0.50–0.60, whereas annual and pasture/fallow parcels drop to 0.31–0.44; bare-soil index tells the inverse story. Their distributions still overlap substantially, so a perennial parcel is not always identifiable, but the average signal is clear.

3.2 Results and model selection

Pooled spatial CV over 44,022 validation parcels (majority baseline macro-F1 0.257):

model	pooled macro-F1	accuracy	PERENNIAL F1
LTAE	0.652	0.691	0.656
PSE-LTAE	0.650	0.681	0.673
LightGBM	0.650	0.701	0.683
<i>rules</i> (transparent phenology control)	0.395	0.367	0.493



Regrouping raises macro-F1 from 0.427 to 0.652. The ML models are effectively tied, so **LightGBM was selected**: it is roughly 100× cheaper on the 215k parcel-year panel and has the best accuracy and perennial F1. Its locked test set was used once, yielding macro-F1 **0.681**, accuracy **0.748**, and class F1 scores of 0.833 (**ANNUAL**), 0.769 (**PERENNIAL**), and 0.440 (**PASTURE_FALLOW**). **PERENNIAL** ROC AUC is **0.937**.

3.3 The **rules** control

The fourth model is a transparent phenology control rather than a learned model. It encodes the expected pattern: perennials have a high NDVI floor and low seasonal amplitude; annual crops have a stronger green peak and/or amplitude; the rest are pasture/fallow.

```
if NDVI_p25 >= t_hi and NDVI_amp <= t_amp: PERENNIAL
elif NDVI_amp > t_amp or NDVI_max >= t_peak: ANNUAL
else: PASTURE_FALLOW
```

Only the three thresholds were fitted within each training fold. The rule reaches macro-F1 0.395—above the majority baseline but well below the ML models. Its thresholds are stable across folds, which makes it a useful scientific comparison, but it is too simple for panel inference.

Replacing **NDVI_max** with **BSI_max** improves the rule's pooled macro-F1 to 0.451 and produces a more plausible class mix, but lowers perennial F1 (0.493 to 0.379) and yields less stable thresholds. It remains an exploratory result, not the adopted model. The broader practical lesson is that dry-season and bare-soil features may be useful future improvements.

4. The panel — applying a single-year model to 28 years

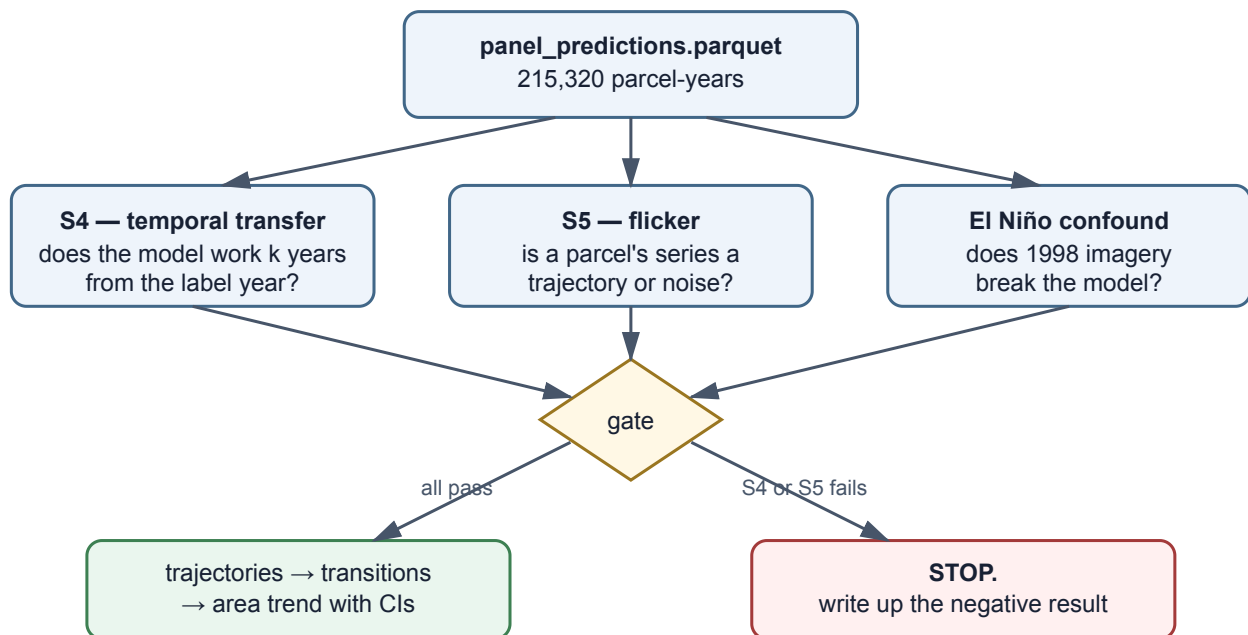
The panel contains 7,690 parcels × 28 years (1996–2023): **215,320 parcel-years** and 16.26M pixel observations extracted from GEE and assembled into an isolated feature bundle per year.

It uses Landsat 5 and 7 only. The training data contain essentially no Landsat 8/9 imagery, so including OLI would introduce a sensor change in the period where an export-crop expansion might be expected. The panel starts in 1996 because earlier clear Landsat coverage is too sparse.

Four thin-coverage years are permanently flagged, not interpolated: 1997 (48.3% of parcels pass the coverage gate), 2009 (49.2%), 2011 (43.3%), and 2012 (83.6%).

5. Validation gates

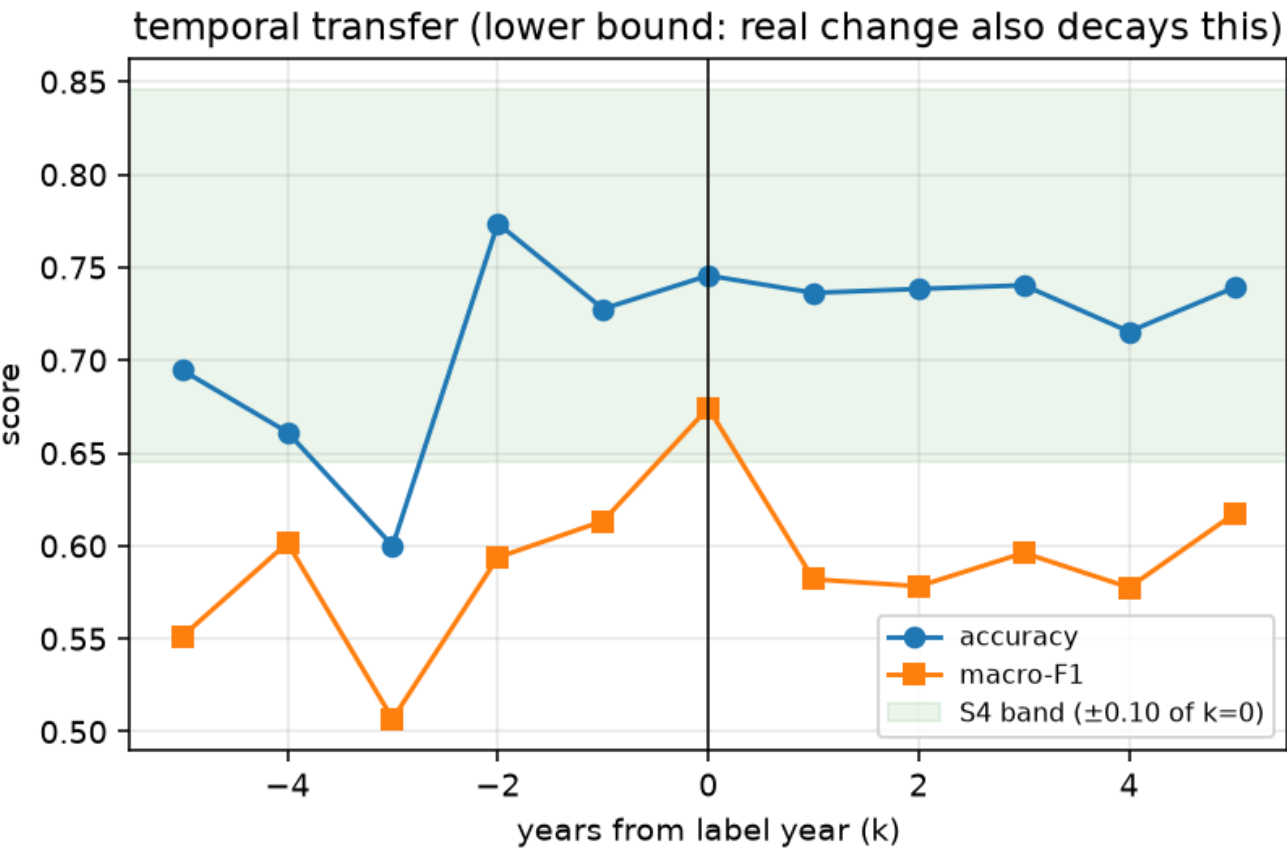
Before a trend could be interpreted, panel predictions had to pass two gates.



S4 — temporal transfer (*gate*)

For each locked-test parcel, predictions at **label_year + k** are scored against its known label for $k = -5$ to $+5$. The criterion is accuracy at $|k| \leq 3$ within 0.10 of label-year accuracy. It asks whether a perennial labelled in 1998 is still recognized in nearby years.

Result: FAIL under both architectures.



k	-5	-4	-3	-2	-1	0	+1	+2	+3
accuracy	.695	.661	.600	.774	.728	.746	.736	.738	.740

Forward transfer is good: +1 to +3 are within 0.010 of the label-year result. The failure is at -3, a bin dominated by the 1996–97 panel years; 1997 alone has accuracy 0.326. The El Niño years are therefore the key failure. **PASTURE_FALLOW** recall also falls away from the label year, though that state is genuinely transient, so S4 is a conservative measure of stability.

S5 — flicker (*gate*)

Each parcel’s 28-year sequence is assessed for class changes. A parcel flickers when it changes class more than **n_observed / 5** times; real land-use conversion should occur only once or twice over this period. The criterion is fewer than 15% flickering **PERENNIAL**-labelled parcels.

Result: FAIL, badly.

PETT label	raw	smoothed	flagged years dropped
ANNUAL	0.210	0.078	0.203
PASTURE_FALLOW	0.583	0.265	0.584
PERENNIAL	0.428	0.221	0.421

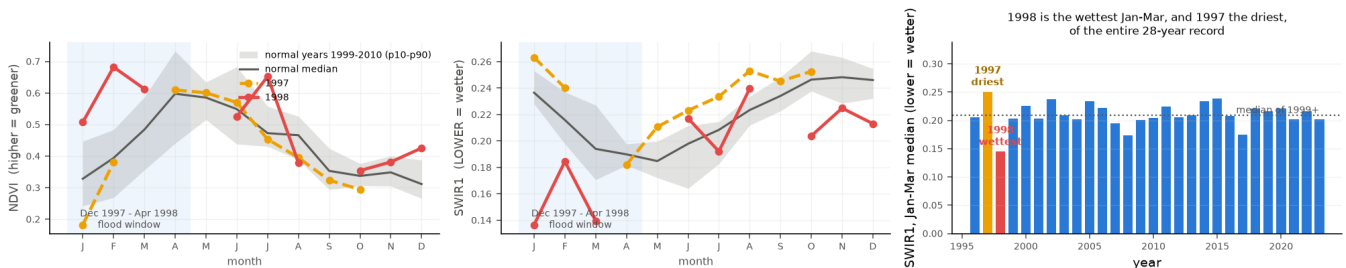
The perennial rate is 0.428 against a 0.15 criterion. Removing thin years changes it little; smoothing halves it but still fails. A persistence rule would hide noise rather than prove a trajectory is real.

6. The El Niño confound — why the baseline fails

6.1 What happened

Piura's 1997–98 El Niño brought exceptional rainfall, flooding, and widespread greening from December 1997 through April 1998. In the imagery, 1998 is unusually wet and green, while 1997 is unusually dry.

The 1997–98 El Niño in the model's own inputs (balanced parcel set; months with < 300 observed parcels dropped)
Jan-Mar 1998 is far wetter (SWIR1 ~0.14 vs ~0.23) AND far greener (NDVI ~0.51-0.61 vs ~0.20-0.46) than any other year — the flooded desert blooms.



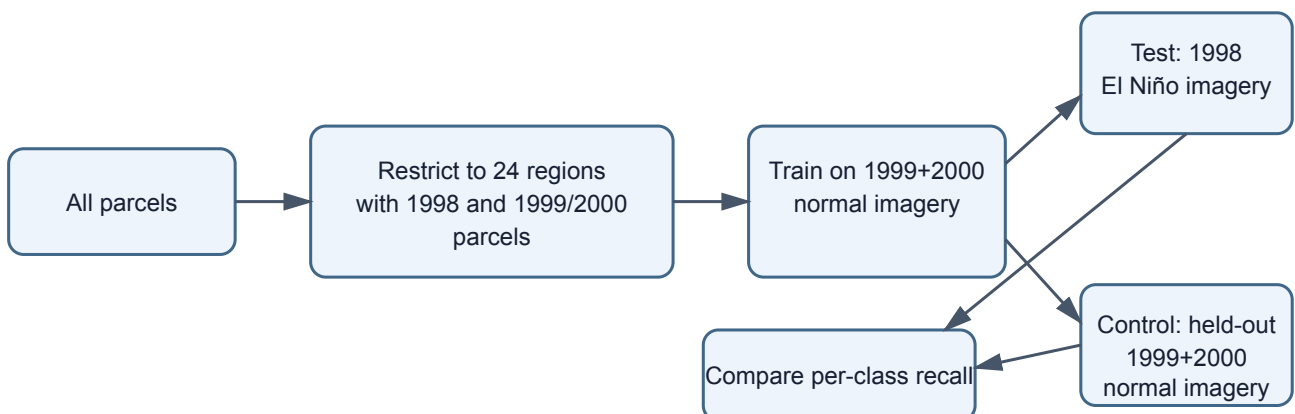
Panel-wide January–March NDVI is 0.51–0.61 in 1998, compared with normal-year medians around 0.33–0.45; SWIR1, a moisture proxy, falls to about 0.14 versus 0.21–0.24 in other years. The two most extreme years in the full record are adjacent and coincide with the concentrated training-label period.

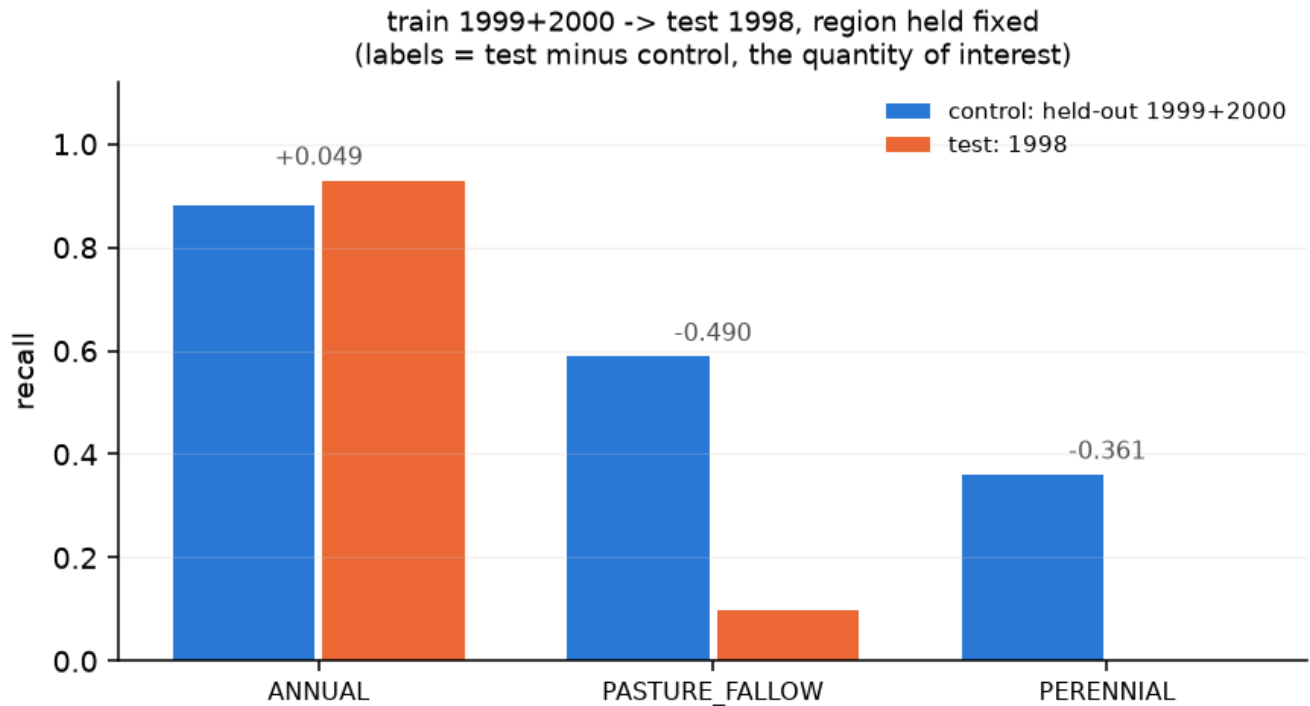
6.2 Why it matters

Around 80% of labels come from 1998–99. In addition, 1998 parcels are just 0.4% perennial because the titling campaign reached rice-growing regions that year. Thus abnormal 1998 imagery is correlated with “not perennial” for reasons unrelated to true land state. Since 1996–98 are also the prospective trend baseline, that confounding directly affects the project’s outcome.

6.3 Controlled test

The key test asks whether the failure comes from 1998 imagery rather than an intrinsically hard 1998 parcel cohort. It restricts analysis to 24 regions containing both 1998 and 1999/2000 parcels, trains on 1999–2000 imagery (4,996 parcels), then compares held-out 1998 parcels with held-out 1999/2000 parcels from the same regions. Per-class recall is used because it is not distorted by the very annual-heavy 1998 class mix.





class	LightGBM control	LightGBM 1998	Δ	LTAE control	LTAE 1998	Δ
ANNUAL	0.833	0.908	+0.076	0.774	0.342	-0.432
PASTURE_FALLOW	0.626	0.142	-0.484	0.632	0.501	-0.130
PERENNIAL	0.544	0.016	-0.527	0.660	0.033	-0.627

Both models collapse on the same 1998 perennial task: LightGBM identifies 54% of perennial parcels in the normal-year control but only 1.6% in the matched 1998 test; LTAE falls from 66% to 3.3%. The comparison holds region and model fixed, so the evidence points to anomalous imagery, not an unusually difficult parcel cohort.

6.4 Why the perennial signal collapses

Perennials are identified largely by a higher NDVI floor and a flatter seasonal profile than annual neighbours. El Niño narrows both contrasts: annual parcels stay greener for longer, while perennial parcels look less consistently green.

Why PERENNIAL recall collapses on 1998 imagery: the contrast that identifies a perennial disappears
Same regions, same held-out parcels. Dot = class median, bar = inter-quartile range. NDVI floor and seasonal swing both stop separating the classes; only BSI survives.



discriminating feature	gap in normal years	gap in 1998
NDVI_p25 — stays green	+1.47 SD	+0.46 SD
NDVI_amp — seasonal senescence	−0.60 SD	−0.10 SD
BSI_max — soil exposure	−0.76 SD	−0.90 SD

The strongest signal, NDVI floor, loses most of its separation, and seasonal amplitude nearly disappears. Bare-soil exposure is more robust, but it is not enough to rescue the trained models. Reweighting toward those features is a hypothesis for later work, not a demonstrated solution.

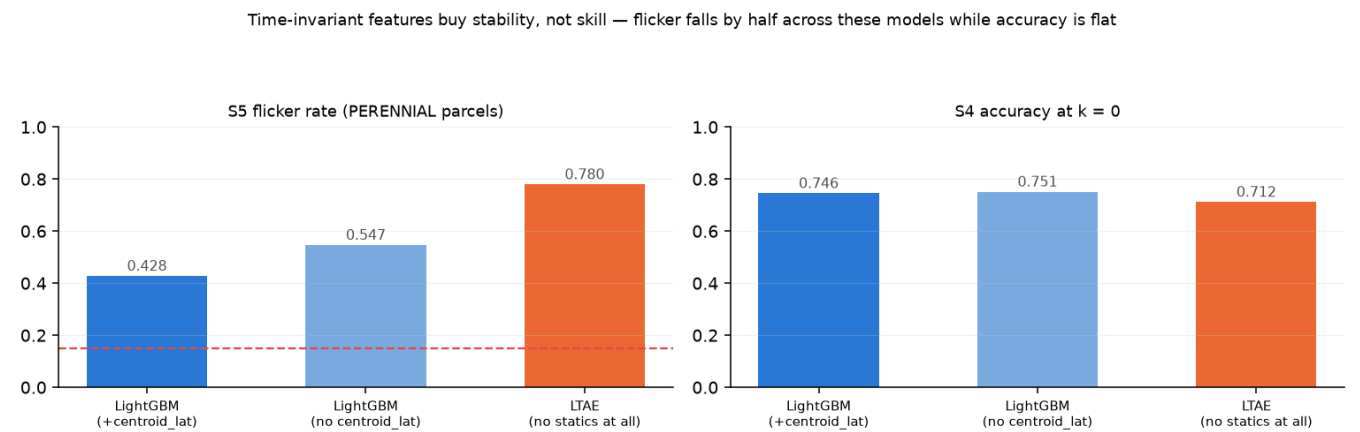
Implication: the apparent early perennial baseline is near zero for artefactual reasons. Any rise from 1996–98 could be an El Niño effect rather than crop conversion. A single early labelled cohort also cannot cleanly separate a general year effect from an El Niño-specific one.

7. Architecture cross-check — no model is currently adequate

The full panel was also inferred with LTAE, which uses per-date sequences rather than whole-year summaries and has no static features. It fails both gates more severely than LightGBM.

	LightGBM nometa	LTAE tuned
S4 accuracy at k = 0	0.746	0.712
S4 worst deviation	0.146 FAIL	0.116 FAIL
S5 flicker, PERENNIAL	0.428 FAIL	0.780 FAIL
S5 flicker, all parcels	0.322	0.842

7.1 Static features can damp apparent flicker



Adding time-invariant information makes a trajectory mechanically flatter while changing single-year accuracy little: a feature that does not vary by year cannot generate a class change. Latitude alone accounts for 23% of LightGBM feature importance. This means a low flicker score can be artificially improved by relying on location rather than current land cover.

The two architectures therefore bracket the issue: LightGBM's static information damps some flicker, while LTAE exposes how unstable predictions based on per-year spectral data can be.

7.2 Conclusion

Per-year 30 m Landsat summaries over parcels averaging about 0.5 ha (roughly five pixels) do not reliably determine the three-class state well enough to support parcel-level annual trajectories. An accuracy near 0.70–0.75 is useful for a single year but insufficient for 28 consecutive parcel-level calls. Trying a substantially different architecture is now a closed route.

8. What survives, and what to do next

What survives

- **The single-year classifier is sound:** locked-test macro-F1 0.681, **PERENNIAL** F1 0.769, ROC AUC 0.937.
- **Forward transfer is good** over +1 to +3 years from the labelled date.
- **The perennial signal is the most robust part** of the three-class analysis, even though it fails in the El Niño baseline.
- **A parcel-level model is needed:** MapBiomass Peru does not assign a perennial-crop code in Piura across 1.97M parcel-years, so it cannot answer this question off the shelf.

What does not survive

The current data do not support per-parcel trajectories, conversion dates, or any area trend anchored on 1996–98. The downstream transition and trend code remains unrun.

Next steps

1. **run for the whole of Peru, not just Piura.** See if there are more years across regions, and see if Piura is hard to infer and the model works well for the rest of Peru.
2. **Change the estimand.** Compare perennial shares across two well-supported windows, e.g. 2000 to 2020 and look at the change on the same parcels, rather than infer annual conversion trajectories. This avoids S5 but not the baseline issue.
3. **Get a second supervision point.** Use the 2012 CENAGRO agricultural census. cant directly link to parcels but would be a good sanity check. can look at farmers who only own one parcel in PETT and assume the same parcel
4. **Drop the El Niño baseline.** Retrain for ≥ 1999 only - lose 80% of the data but 36,300 remain.
5. **Manually annotate 2020 images.** use as train and test data to improve temporal generalisability